

Module 1

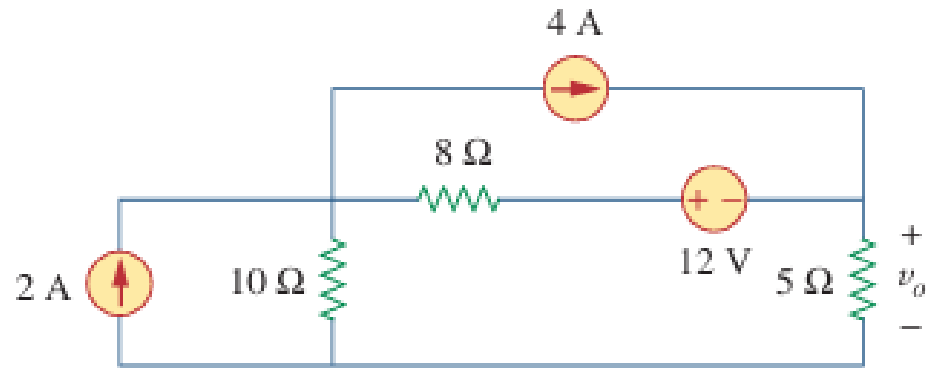
DC Circuit Analysis and Theorems

Dr. Elizabeth Rufus

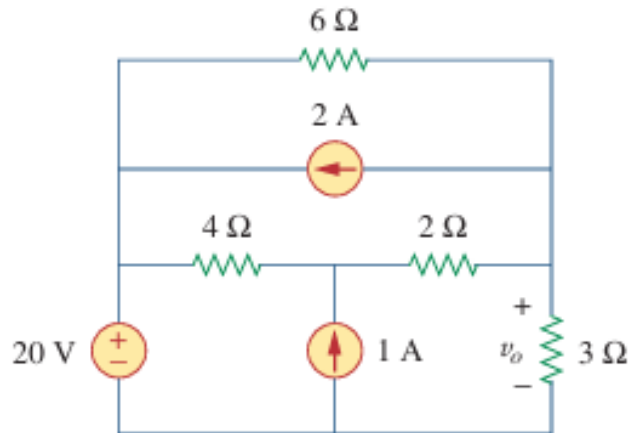
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Tutorial

1. Using Superposition theorem find v_o .

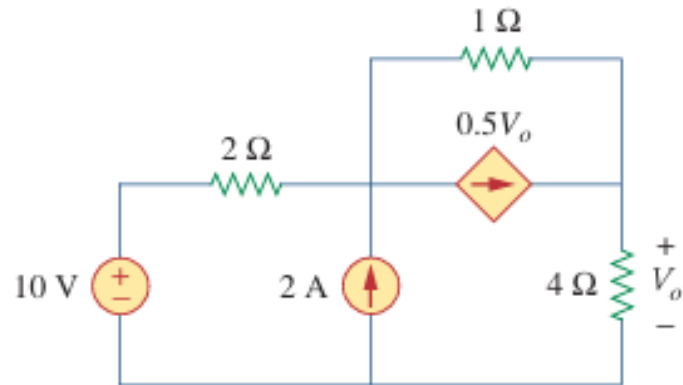


2. Apply the superposition principle to find v_o circuit .

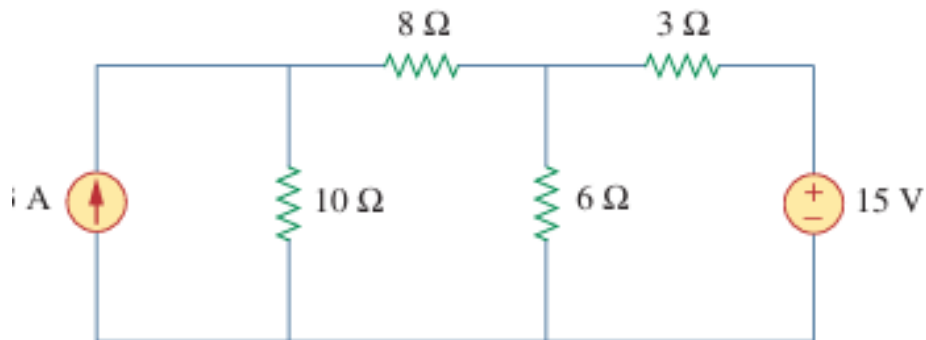


Tutorial

3. Use Superposition to find in the circuit.

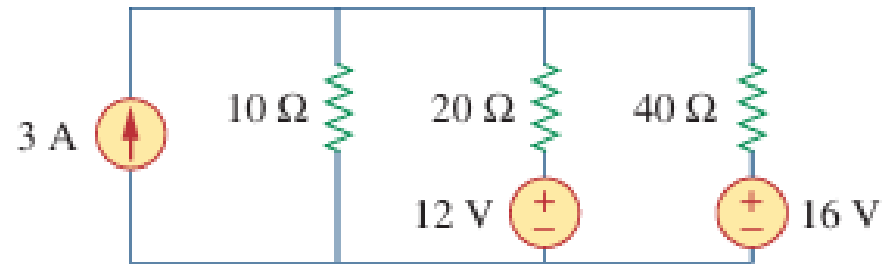


4. Referring to the circuit below, use source transformation to determine the current and power absorbed by the 8 Ω resistor..

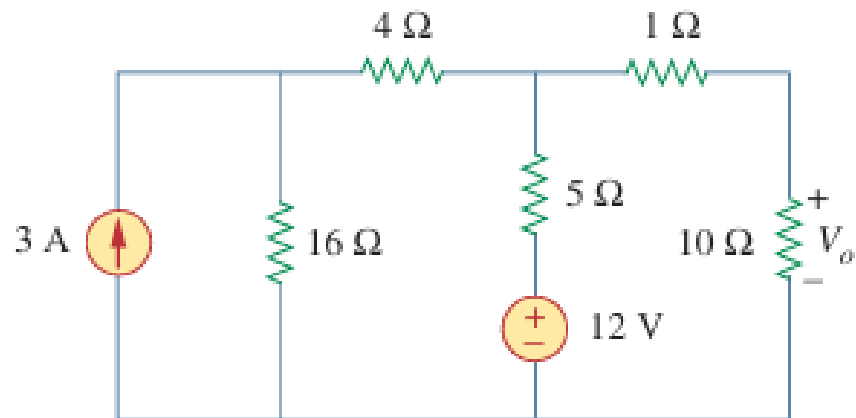


Tutorial

5. Use source transformation to reduce the circuit below into a single voltage source in series with a single resistor.

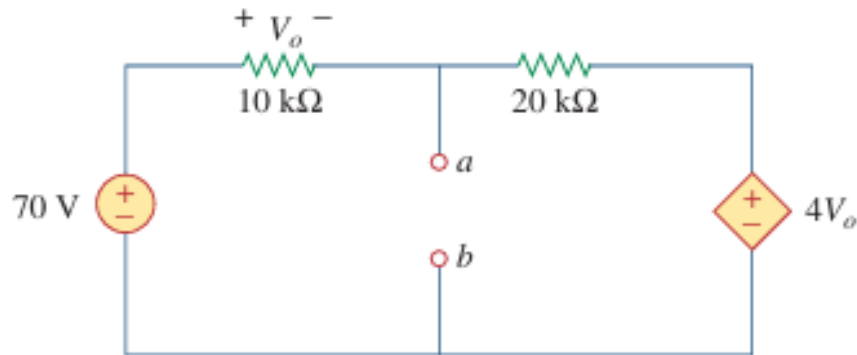


6. Apply Thevenin's theorem to find V_o in the circuit.

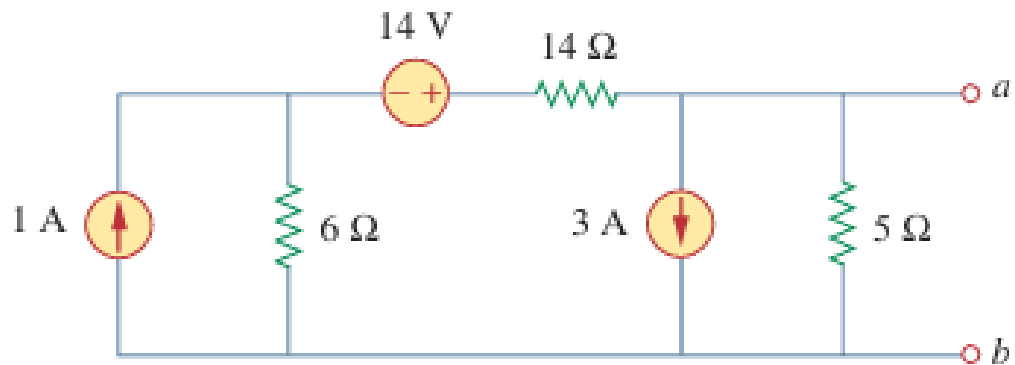


Tutorial

7. Find the Thevenin's equivalent at terminals a-b of the circuit.

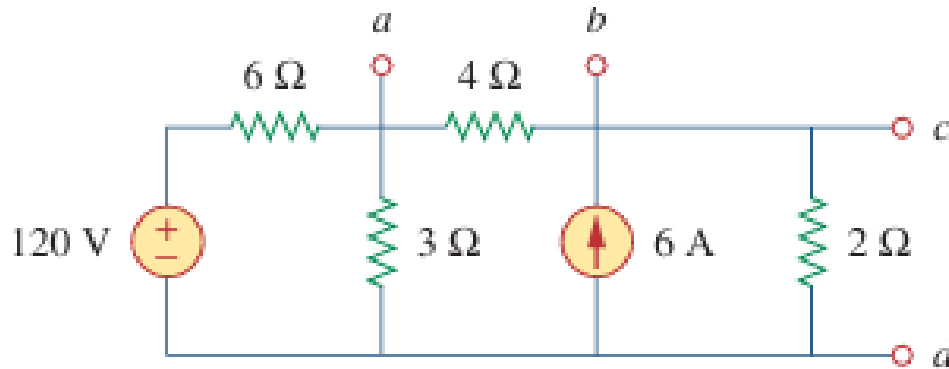


8. Find the Thevenin and Norton equivalents at terminals of the circuit.



Tutorial

9. Given the circuit, obtain the Norton equivalent as viewed from terminals i) a-b ii) c-d.



10. Compute the value of R that results in maximum power transfer to the $10\text{-}\Omega$ resistor in circuit. Find the maximum power.

